

EXPLOSIVE DIGIT DYNAMICS, HYPERGROWTH CASCADES AND INFINITE ATTRACTOR BASINS: ARCHITECTURE OF A NEW INTEGER UNIVERSE.

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Abstract

We investigate a previously undocumented integer transformation whose explosive behavior places it beyond the known extremes of Collatz-type dynamics. The system operates by (1) squaring each digit of an integer and concatenating the results, then (2) repeatedly compressing the expanded sequence by summing every consecutive triplet of digits whenever the total length remains divisible by three. This deceptively elementary process generates combinatorial shockwaves: numerical structures routinely balloon to hundreds or thousands of digits before undergoing catastrophic collapse into a microscopic attractor set.

[illegible]

The system displays deterministic chaos, hyperinflationary digit cascades, symmetry-breaking, digit-mass dissipation and stability phenomena normally reserved for nonlinear differential systems, not simple base-10 arithmetic. The existence of violent growth-collapse cycles coupled with guaranteed stabilization suggests a previously unknown class of “digit-dynamical attractor systems” with measurable entropy flow, invariant fixed points and universality across the entire tested integer domain.

Its structural sensitivity and long-range determinism hint at potential applications in cryptographically hard-to-predict transforms, entropy amplification, pseudo-random sequence generation, high-dimensional pattern analysis and the study of discrete dissipative systems. This work formalizes the transformation, documents its empirical invariants and presents that a new mathematically rich dynamical regime exists within elementary arithmetic, one that may stand alongside Collatz and Kaprekar as a landmark integer phenomenon of the 21st century.

1.Introduction

The transformation studied in this paper begins with nothing more exotic than squaring digits and grouping them in threes yet the resulting behavior is so violently nonlinear that even the modest integers erupt into multi hundred-digit monsters before collapsing into microscopic fixed points.

This process did not emerge from classical number theory, probabilistic modelling or dynamical systems research. It emerged from a raw, structural intuition: that digits contain latent combinatorial energy and that squaring, concatenating and compressing them in specific sequences may reveal hidden skeletons inside integer arithmetic.

Most well-known integer maps have a single signature trait:

Collatz: unpredictability wanders but values remain small.

Kaprekar routines: reorder digits but never explode uncontrollably.

Conway FRACTRAN: explosive but defined over fractions, not digits.

Our system combines the most extreme behaviors of all:

[1] Hyper explosive growth (worse than Collatz or Kaprekar by orders of magnitude)

[2] Absolute-deterministic convergence (unlike chaos systems)

[3] Multiple fixed-point attractors (unlike Collatz's single cycle hypothesis)

[4] Digit level physics (a totally new regime)

This puts the transformation in its own class: a violent digit dynamic system with guaranteed stabilization.

It is rare in mathematics for a process to be simple, computationally explosive, empirically universal, convergent across all tests and structurally rich enough to generate attractors.

Systems with this combination of traits are historically rare. When they appear, Collatz (1937), Kaprekar (1949) and Conway's FRACTRAN (1987), they generate decades of interest, cross disciplinary analysis and deep open questions.

Our system volatile growth phase is far more dramatic than any of the classics.

2. Illustrations

2.1 The number 1000081

This number takes 109 steps to reach the stable number 10 as illustrated below:

Enter the starting number: 1,000,081

Step 1 (current number): 1,000,081

-> After squaring digits: 116000160641

-> After summing groups of 3: 80711

Step 2 (current number): 80711

-> After squaring digits: 6404911

Step 3 (current number): 6404911

-> After squaring digits: 36160168111

Step 4 (current number): 36160168111

-> After squaring digits: 936136013664111

-> After summing groups of 3: 18104163

Step 5 (current number): 18104163

-> After squaring digits: 16410161369

Step 6 (current number): 16410161369

-> After squaring digits: 1361610136193681

Step 7 (current number): 1361610136193681

-> After squaring digits: 1936136101936181936641

Step 8 (current number): 1936136101936181936641

-> After squaring digits: 18193619361018193616418193636161

Step 9 (current number): 18193619361018193616418193636161

-> After squaring digits: 16418193618193610164181936136161641819369361361

Step 10 (current number): 16418193618193610164181936136161641819369361361

-> After squaring digits:

1361616418193616418193610136161641819361936136136161641819368193619361

Step 11 (current number):

1361616418193616418193610136161641819361936136136161641819368193619361

-> After squaring digits:

1936136136161641819361361616418193610193613613616164181936181936193619361361
3616164181936641819361819361

Step 12 (current number):

1936136136161641819361361616418193610193613613616164181936181936193619361361
3616164181936641819361819361

-> After squaring digits:

1819361936193613613616164181936193613613616164181936101819361936193613613616
1641819361641819361819361819361936193613613616164181936361616418193616418193
61

Step 13 (current number):

1819361936193613613616164181936193613613616164181936101819361936193613613616
1641819361641819361819361819361936193613613616164181936361616418193616418193
61

-> After squaring digits:

1641819361819361819361936193613613616164181936181936193619361361361616418193
6101641819361819361819361936193613613616164181936136161641819361641819361641
8193618193618193619361936136136161641819369361361361616418193613616164181936
1

Step 14 (current number):

1641819361819361819361936193613613616164181936181936193619361361361616418193
6101641819361819361819361936193613613616164181936136161641819361641819361641
8193618193618193619361936136136161641819369361361361616418193613616164181936
1

-> After squaring digits:

1361616418193616418193616418193618193618193619361936136136161641819361641819
3618193618193619361936136136161641819361013616164181936164181936164181936181
9361819361936193613613616164181936193613613616164181936136161641819361361616
4181936164181936164181936181936181936193619361361361616418193681936193619361
36136161641819361936136136161641819361

-> After summing groups of 3:

1081118101118101118101810181018131010131110181110181018101813161010101313137
1081118101118101118101810181018131010131110181310101311101810811181010131313
131313131313151315131610181010811181713161018101081118101810108111810

Step 15 (current number):

1081118101118101118101810181018131010131110181110181018101813161010101313137
1081118101118101118101810181018131010131110181310101311101810811181010131313
131313131313151315131610181010811181713161018101081118101810108111810

-> After squaring digits:

1064111641011164101116410164101641016419101019111016411101641016410164191361
0101019191949106411164101116410111641016410164101641910101911101641910101911
10164106411164101019191919191919191251912519136101641010641116414919136101
6410106411164101641010641116410

Step 16 (current number):

1064111641011164101116410164101641016419101019111016411101641016410164191361
0101019191949106411164101116410111641016410164101641910101911101641910101911
10164106411164101019191919191919191251912519136101641010641116414919136101
6410106411164101641010641116410

-> After squaring digits:

1036161113616101113616101113616101361610136161013616181101018111101361611101
3616101361610136161811936101010181181181168110361611136161011136161011136161
0136161013616101361618110101811110136161811010181111013616103616111361610101
8118118118118118118118114251811425181193610136161010361611136161168118119
361013616101036161113616101361610103616111361610

Step 17 (current number):

1036161113616101113616101113616101361610136161013616181101018111101361611101
3616101361610136161811936101010181181181168110361611136161011136161011136161
0136161013616101361618110101811110136161811010181111013616103616111361610101
8118118118118118118118114251811425181193610136161010361611136161168118119
361013616101036161113616101361610103616111361610

-> After squaring digits:

1093613611193613610111936136101119361361019361361019361361019361361641101016
4111101936136111019361361019361361019361361641181936101010164116411641136641
1093613611193613610111936136101119361361019361361019361361019361361641101016
4111101936136164110101641111019361361093613611193613610101641164116411641164
1164116411641164111642516411164251641181936101936136101093613611193613611366
4116411819361019361361010936136111936136101936136101093613611193613610

-> After summing groups of 3:

1010101110102131073181021810218102181011221131010102131071310713101361810121
111861561210813107318102111010101010101010101127621310136111321810101010
1110101118611118611118611811113121261810101010118103181051686181010101011810
318102181021210813107

-> After summing groups of 3:

2131681231039112552168584101033151249411942121212215621051121222108108108105
49921104106201012291231033128

-> After summing groups of 3: 6156411129215497141454513645399918125733126611

Step 18 (current number): 6156411129215497141454513645399918125733126611

-> After squaring digits:

36125361611148141251681491161162516251936162598181811641425499914363611

Step 19 (current number):

36125361611148141251681491161162516251936162598181811641425499914363611

-> After squaring digits:

9361425936136111166411614251366411681113611364251364251819361364258164164164
1136161164251681818111693693611

Step 20 (current number):

9361425936136111166411614251366411681113611364251364251819361364258164164164
1136161164251681818111693693611

-> After squaring digits:

8193611642581936193611113636161136116425193636161136641119361193616425193616
4251641819361936164256413616136161361611936136113616425136641641641113681936
8193611

-> After summing groups of 3:

1810111513161031213581215121351631811101215101212131316101215108101013111010
101181511116101817138

Step 21 (current number):

1810111513161031213581215121351631811101215101212131316101215108101013111010
101181511116101817138

-> After squaring digits:

1641011125191361091419256414125141925136916411101412510141419191361014125106
4101019111010101164125111136101641491964

Step 22 (current number):

1641011125191361091419256414125141925136916411101412510141419191361014125106
4101019111010101164125111136101641491964

-> After squaring digits:

1361610111425181193610811161814253616116142511618142519368113616111011614251
0116116181181193610116142510361610101811110101011361614251111936101361611681
1813616

-> After summing groups of 3:

1082714111098101110811881381810108281128810101828114132932151377111041381012
13

-> After summing groups of 3: 91061093917121011841711110814811951226

Step 23 (current number): 91061093917121011841711110814811951226

-> After squaring digits: 8110361081981149141011641614911110641166411812514436

Step 24 (current number): 8110361081981149141011641614911110641166411812514436

-> After squaring digits:

641109361064181641116811161011361613611681111103616113636161164142511616936

-> After summing groups of 3: 1110101010118108210108153413515811771318

Step 25 (current number): 1110101010118108210108153413515811771318

-> After squaring digits: 111010101011641064410106412591619251256411494919164

-> After summing groups of 3: 3122111057715168136171911

Step 26 (current number): 3122111057715168136171911

-> After squaring digits: 9144111025494912513664193614918111

Step 27 (current number): 9144111025494912513664193614918111

-> After squaring digits: 811161611104251681168114251936361618193611681164111

-> After summing groups of 3: 10885815156818101513815113

Step 28 (current number): 10885815156818101513815113

-> After squaring digits: 10646425641251253664164101251964125119

Step 29 (current number): 10646425641251253664164101251964125119

-> After squaring digits:

1036163616425361614251425936361613616101425181361614251181

Step 30 (current number):

1036163616425361614251425936361613616101425181361614251181

-> After squaring digits:

1093613693613616425936136116425116425819369361361936136101164251641936136116
42511641

-> After summing groups of 3: 10101810101217108127121418181013107811111310812711

Step 31 (current number): 10101810101217108127121418181013107811111310812711

-> After squaring digits:

1010164101014149106414491411616416410191049641111191064144911

Step 32 (current number):

1010164101014149106414491411616416410191049641111191064144911

-> After squaring digits:

1010136161010116116811036161161681116113613616136161018110168136161111181103
616116168111

Step 33 (current number):

1010136161010116116811036161161681116113613616136161018110168136161111181103
616116168111

-> After squaring digits:

1010193613610101136113664110936136113613664111361193619361361936136101641101
366419361361111164110936136113613664111

Step 34 (current number):

1010193613610101136113664110936136113613664111361193619361361936136101641101
366419361361111164110936136113613664111

-> After squaring digits:

1010181936193610101193611936361611081936193611936193636161119361181936181936
1936181936193610136161101936361618193619361111136161108193619361193619363616
111

Step 35 (current number):

1010181936193610101193611936361611081936193611936193636161119361181936181936
1936181936193610136161101936361618193619361111136161108193619361193619363616
111

-> After squaring digits:

1010164181936181936101011819361181936936136110641819361819361181936181936936
1361118193611641819361641819361819361641819361819361019361361101819369361361
6418193618193611111936136110641819361819361181936181936936136111

-> After summing groups of 3:

2713131513721018101318108101018101810131513181081018813131313131513131313151
371310891318101313131513831810211181018101018101818103

Step 36 (current number):

2713131513721018101318108101018101810131513181081018813131313131513131313151
371310891318101313131513831810211181018101018101818103

-> After squaring digits:

4491919125194941016410191641064101016410164101912519164106410164641919191919
12519191912519491910648119164101919191251964916410411164101641010164101641
64109

Step 37 (current number):

4491919125194941016410191641064101016410164101912519164106410164641919191919
12519191912519491910648119164101919191251964916410411164101641010164101641
64109

-> After squaring digits:

1616811811811425181168116101361610181136161036161010136161013616101811425181
1361610361610136163616181181181181181142518118118114251811681181103616641
1811361610181181181142518136168113616101611136161013616101013616101361613616
1081

Step 38 (current number):

1616811811811425181168116101361610181136161036161010136161013616101811425181
1361610361610136163616181181181181181142518118118114251811681181103616641
1811361610181181181142518136168113616101611136161013616101013616101361613616
1081

-> After squaring digits:

1361366411641164111642516411366411361019361361016411936136109361361010193613
6101936136101641116425164119361361093613610193613693613616411641164116411641
1164251641164116411641116425164113664116411093613636161164119361361016411641
1641116425164193613664119361361013611193613610193613610101936136101936136193
613610641

Step 39 (current number):

1361366411641164111642516411366411361019361361016411936136109361361010193613
6101936136101641116425164119361361093613610193613693613616411641164116411641
1164251641164116411641116425164113664116411093613636161164119361361016411641
1641116425164193613664119361361013611193613610193613610101936136101936136193
613610641

-> After squaring digits:

1936193636161136161136161113616425136161193636161193610181936193610136161181
9361936108193619361010181936193610181936193610136161113616425136161181936193
6108193619361018193619368193619361361611361611361611361611136164251361
6113616113616113616111361642513616119363616113616110819361936936136113616118
1936193610136161136161136161113616425136161819361936361611819361936101936111
8193619361018193619361010181936193610181936193618193619361036161

-> After summing groups of 3:

1316121351351331012913111213111091316104131013161091813721810182181018210810
1181081810189131610913161713161010810810810810118108108108101181081810
8108918131810810818101821081081081011810151316121310131610101010131610913161
0110181371018131513161098

Step 40 (current number):

1316121351351331012913111213111091316104131013161091813721810182181018210810
1181081810189131610913161713161010810810810810118108108108101181081810
8108918131810810818101821081081081011810151316121310131610101010131610913161
0110181371018131513161098

-> After squaring digits:

1913614192519251991014811911141911108119136101619101913610811641949416410164
4164101644106410116410641641016481191361081191361491913610106410641064106410
6410116410641064106410641011641064164106410648116419164106410641641016441064
1064106410116410125191361419101913610101010191361081191361011016419491016419
12519136108164

-> After summing groups of 3:

1110148161921311611210137810111091114141179117910211711511101371013111910210
7511107585111075112111011751881411751111791075112113151061011101210109111022
1122211121510911

-> After summing groups of 3:

35151258441621039911173972112511312312142134291413979101345673321025544611

Step 41 (current number):

35151258441621039911173972112511312312142134291413979101345673321025544611

-> After squaring digits:

9251251425641616136410981811114998149411425119149141164191648111619814981101
916253649994104252516163611

Step 42 (current number):

9251251425641616136410981811114998149411425119149141164191648111619814981101
916253649994104252516163611

-> After squaring digits:

8142514251164253616136136193616108164164111116818164116811611164251181116811
1611361618113616641113618164116816411018113642593616818181161016425425136136
93611

Step 43 (current number):

8142514251164253616136136193616108164164111116818164116811611164251181116811
1611361618113616641113618164116816411018113642593616818181161016425425136136
93611

-> After squaring digits:

6411642511642511361642593613619361936181936136106413616136161111136641641361
6113664113611136164251164111366411136119361361641193613636161119361641361611
366413616110164119361642581936136641641641136101361642516425193619368193611

Step 44 (current number):

6411642511642511361642593613619361936181936136106413616136161111136641641361
6113664113611136164251164111366411136119361361641193613636161119361641361611
366413616110164119361642581936136641641641136101361642516425193619368193611

-> After squaring digits:

3616113616425113616425119361361642581936193618193618193616418193619361036161
9361361936136111119363616136161936136119363616119361119361361642511361611193
6361611193611819361936136161181936193693613611181936136161936136119363616193
6136110136161181936136164256418193619363616136161361611936101936136164251361
64251819361819366418193611

-> After summing groups of 3:

1081012710127181011151316101810181011181018413131016101031315810161010131581
3813101311513318108138181018108101813181081018108181011121313108413101310131
11118101810101310813713101311101181810181513138

Step 45 (current number):

1081012710127181011151316101810181011181018413131016101031315810161010131581
3813101311513318108138181018108101813181081018108181011121313108413101310131
11118101810101310813713101311101181810181513138

-> After squaring digits:

1064101449101449164101112519136101641016410111641016416191910136101091912564
1013610101912564196419101911251991641064196416410164106410164191641064101641
0641641011141919106416191019101911111641016410101910641949191019111011641641
0164125191964

Step 46 (current number):

1064101449101449164101112519136101641016410111641016416191910136101091912564
1013610101912564196419101911251991641064196416410164106410164191641064101641
0641641011141919106416191019101911111641016410101910641949191019111011641641
0164125191964

-> After squaring digits:

1036161011616811011616811361610111425181193610136161013616101113616101361613
6181181101936101081181142536161019361010181142536161813616181101811142518181
1361610361618136161361610136161036161013616181136161036161013616103616136161
0111161811811036161361811018110181111113616101361610101811036161811681181101
81111011361613616101361614251811813616

-> After summing groups of 3:

41328152815108271411104132107310710810101010101010110713721011101510810931
1101010710151081071089841310108984134131082810104131010210933107108110413101
51021032101013210118101213

Step 47 (current number):

41328152815108271411104132107310710810101010101010110713721011101510810931
1101010710151081071089841310108984134131082810104131010210933107108110413101
51021032101013210118101213

-> After squaring digits:

161946412546412510644491161110161941049910491064101010101010101110491949410
1110125106410819111010104910125106410491064816416191010648164161916191064464
10101619101041081991049106411016191012510410941010194101164101419

Step 48 (current number):

161946412546412510644491161110161941049910491064101010101010101110491949410
1110125106410819111010104910125106410491064816416191010648164161916191064464
10101619101041081991049106411016191012510410941010194101164101419

-> After squaring digits:

1361811636161425163616142510361616168111361110136181161016818110168110361610
1010101011010111016811811681161011101425103616106418111101010168110142510361
6101681103616641361613618110103616641361613618113618110361616361610101361811
0101610641818110168110361611013618110142510161081161010181161011361610116181

Step 49 (current number):

1361811636161425163616142510361616168111361110136181161016818110168110361610
1010101011010111016811811681161011101425103616106418111101010168110142510361
6101681103616641361613618110103616641361613618113618110361616361610101361811
0101610641818110168110361611013618110142510161081161010181161011361610116181

-> After squaring digits:

1936164113693613611642513693613611642510936136136136641119361110193616411361
013664164110136641109361361010101010101101366411641136641136101110116425
1093613610361616411110101013664110116425109361361013664110936136361619361361
9361641101093613636161936136193616411936164110936136136936136101019361641101
0136103616164164110136641109361361101936164110116425101361064113610101641136
10119361361011361641

Step 50 (current number):

1936164113693613611642513693613611642510936136136136641119361110193616411361
013664164110136641109361361010101010101101366411641136641136101110116425
1093613610361616411110101013664110116425109361361013664110936136361619361361
9361641101093613636161936136193616411936164110936136136936136101019361641101
0136103616164164110136641109361361101936164110116425101361064113610101641136
10119361361011361641

-> After squaring digits:

1819361361611936819361936113616425193681936193611361642510819361936193619363
6161118193611101819361361611936101936361613616110193636161108193619361010101
0101101011101936361611361611936361611936101110113616425108193619361093613613
6161111010101936361611011361642510819361936101936361611081936193693613618193
6193618193613616110108193619369361361819361936181936136161181936136161108193
61936193681936193610101819361361611010193610936136136161101936361611081
9361936110181936136161101136164251019361036161193610101361611936101181936193
61011936136161

-> After summing groups of 3:

1018108131518138101215171316101011891813161018108101839131013111010121310821
8108918137212222181081081810818225131191316101210101332113158210118918137131
5891316181010181018101810829181318101513161018101081810108918131617131610110
1810821010121010131082181089181389131013251311218413111011081371013161021810
8

Step 51 (current number):

1018108131518138101215171316101011891813161018108101839131013111010121310821
8108918137212222181081081810818225131191316101210101332113158210118918137131
5891316181010181018101810829181318101513161018101081810108918131617131610110
1810821010121010131082181089181389131013251311218413111011081371013161021810
8

-> After squaring digits:

1016410641912516419641014125149191361010116481164191361016410641016498119101
9111010141910644164106481164194941444416410641064164106416444251911811913610
1410101994119125644101164811641949191256481191361641010164101641016410644811
6419164101251913610164101064164101064811641913614919136101101641064410101410
1019106441641064811641964811910194251911416416191110110641949101913610416410
64

Step 52 (current number):

1016410641912516419641014125149191361010116481164191361016410641016498119101
9111010141910644164106481164194941444416410641064164106416444251911811913610
1410101994119125644101164811641949191256481191361641010164101641016410644811
6419164101251913610164101064164101064811641913614919136101101641064410101410
1019106441641064811641964811910194251911416416191110110641949101913610416410
64

-> After squaring digits:

1013616103616181142513616181361610116142511681181193610101136166411361618119
3610136161036161013616816411811018111101011618110361616136161036166411361618
1168116116161616136161036161036161361610361613616161642518111641181193610116
1010181811611811425361616101136166411361618116811811425361664118119361361610
1013616101361610136161036161664113616181136161014251811936101361610103616136
1610103616641136161811936116811811936101101361610361616101011610101811036161
6136161036166411361618136166411811018116425181111613616136181111011036161811
681101811936101613616103616

Step 53 (current number):

1013616103616181142513616181361610116142511681181193610101136166411361618119
3610136161036161013616816411811018111101011618110361616136161036166411361618
1168116116161616136161036161036161361610361613616161642518111641181193610116
1010181811611811425361616101136166411361618116811811425361664118119361361610
1013616101361610136161036161664113616181136161014251811936101361610103616136
1610103616641136161811936116811811936101101361610361616101011610101811036161
6136161036166411361618136166411811018116425181111613616136181111011036161811
681101811936101613616103616

-> After squaring digits:

1019361361093613616411164251936136164193613610113611642511366411641181936101
0119361363616119361361641181936101936136109361361019361366413616116411016411
1101011361641109361361361936136109361363616119361361641136641136113613613613
6193613610936136109361361936136109361361936136136136164251641113616116411819
3610113610101641641136116411164259361361361011936136361611936136164113664116
4111642593613636161164118193619361361010193613610193613610193613610936136136
3616119361361641193613610116425164118193610193613610109361361936136101093613
6361611936136164118193611366411641181936101101936136109361361361010113610101
6411093613613619361361093613636161193613616419361363616116411016411361642516
4111136193613619361641111011093613616411366411016411819361013619361361093613
6

Step 54 (current number):

1019361361093613616411164251936136164193613610113611642511366411641181936101
0119361363616119361361641181936101936136109361361019361366413616116411016411
1101011361641109361361361936136109361363616119361361641136641136113613613613
6193613610936136109361361936136109361361936136136136164251641113616116411819
3610113610101641641136116411164259361361361011936136361611936136164113664116
4111642593613636161164118193619361361010193613610193613610193613610936136136
3616119361361641193613610116425164118193610193613610109361361936136101093613
6361611936136164118193611366411641181936101101936136109361361361010113610101
6411093613613619361361093613636161193613616419361363616116411016411361642516
4111136193613619361641111011093613616411366411016411819361013619361361093613
6

-> After squaring digits:

1018193619361081936193613616111361642518193619361361618193619361011936113616
4251193636161136161164181936101011819361936936136118193619361361611641819361
0181936193610819361936101819361936361619361361136161101361611110101193613616
1108193619361936181936193610819361936936136118193619361361611936361611936119
3619361936193618193619361081936193610819361936181936193610819361936181936193
6193619361361642513616111936136113616116418193610119361010136161361611936113
6161113616425819361936193610118193619369361361181936193613616119363616113616
1113616425819361936936136113616116418193618193619361010181936193610181936193
6101819361936108193619361936936136118193619361361611819361936101136164251361
6116418193610181936193610108193619361819361936101081936193693613611819361936
1361611641819361193636161136161164181936101101819361936108193619361936101011
9361010136161108193619361936181936193610819361936936136118193619361361618193

6193693613611361611013616119361361642513616111193618193619361819361361611110
11081936193613616119363616110136161164181936101936181936193610819361936

-> After summing groups of 3:

2181018913161010810118181018108181018213810127181081081118101101316181010101
8131013813137101813718101821810181016101010821082213101321810181315131610918
1318108181018108131581381813161018101813718101891316101810189131610181018131
6101012913318105138131371110110810818513310121418131610218101818101013161010
8181081081011151316181010108111018101813721810182181018218101891316101818101
0131610108181018210118108111810913161011810181018137913161810101018131013813
1381810810811181029131610918131610111101108913161018101813718101818101013161
0101513161810101082108181011810811101810181018108329181310131112132108111018
2181018137181018

-> After summing groups of 3:

1121810292172917211123169939251521104125119111091129819312541121271010101291
0995141212892121621810291857109572127991211329914736128391092102999939892921
0911112119111091389101579109310939138291091113892954121210991031381012822218
102910119109210265152111991093929101112122546921110911109

-> After summing groups of 3:

4101110121041021168277111041813114102112101571112911318131410181252012101310
1211272020311411310202211042112121051918431941231211310111031383191220245151
221110

Step 55 (current number):

4101110121041021168277111041813114102112101571112911318131410181252012101310
1211272020311411310202211042112121051918431941231211310111031383191220245151
221110

-> After squaring digits:

1610111014101610411366444949111016164191116104114101254911148111916419116101
6414254014101910141144940409111611910404411016411414102518116416918116149141
191011109196491811440416251251441110

Step 56 (current number):

1610111014101610411366444949111016164191116104114101254911148111916419116101
6414254014101910141144940409111611910404411016411414102518116416918116149141
191011109196491811440416251251441110

-> After squaring digits:

1361011101161013610161193636161616811681111013613616181111361016111610142516
8111116641118113616181113610136161164251601161018110116111616811601608111136
1181101601616110136161116116104251641136161368116411361168111611811011108118
13616811641116160161364251425116161110

Step 57 (current number):

1361011101161013610161193636161616811681111013613616181111361016111610142516
8111116641118113616181113610136161164251601161018110116111616811601608111136
1181101601616110136161116116104251641136161368116411361168111611811011108118
13616811641116160161364251425116161110

-> After squaring digits:

1936101110113610193610136118193693613613613664113664111101936193613616411119
3610136111361011642513664111113636161116411936136164111936101936136113616425
1360113610164110113611136136641136013606411119361164110136013613611019361361
1136113610164251361611936136193664113616119361136641113611641101110641164193
61366411361611136136013619361642511642511361361110

-> After summing groups of 3:

1373210101048181810101015615621316101011318210578111011310108111110101111101
0101010118957112581010111041261110112104108101010581071110813101615610818516
3101123108141015610810104161012712710102

-> After summing groups of 3:

1161517101212135725111210251103132212181389227824591699957129141026966795831
083

Step 58 (current number):

1161517101212135725111210251103132212181389227824591699957129141026966795831
083

-> After squaring digits:

1136125149101414192549425111410425110919441416419648144496441625811368181812
54914811161043681363649812564910649

-> After summing groups of 3:

591010614111576671017611191317149141017101114108715121911151019

-> After summing groups of 3: 1511131814881111146826913411710

Step 59 (current number): 1511131814881111146826913411710

-> After squaring digits: 1251119164116646411111163664436811916114910

Step 60 (current number): 1251119164116646411111163664436811916114910

-> After squaring digits:

142511181136161136361636161111113693636161693664118113611168110

-> After summing groups of 3: 77101081010158351815818161058152

Step 61 (current number): 77101081010158351815818161058152

-> After squaring digits: 4949101064101012564925164125641641361025641254

Step 62 (current number): 4949101064101012564925164125641641361025641254

-> After squaring digits:

1681168110103616101014253616814251361614253616136161936104253616142516

Step 63 (current number):

1681168110103616101014253616814251361614253616136161936104253616142516

-> After squaring digits:

1366411366411010936136101011642593613664116425193613611642593613619361361819
361016425936136116425136

Step 64 (current number):

1366411366411010936136101011642593613664116425193613611642593613619361361819
361016425936136116425136

-> After squaring digits:

1936361611936361611010819361936101011361642581936193636161136164251819361936
113616425819361936181936193616418193610136164258193619361136164251936

Step 65 (current number):

1936361611936361611010819361936101011361642581936193636161136164251819361936
113616425819361936181936193616418193610136164258193619361136164251936

-> After squaring digits:

1819369361361181936936136110106418193618193610101193613616425641819361819369
3613611936136164251641819361819361193613616425641819361819361641819361819361
361616418193610193613616425641819361819361193613616425181936

Step 66 (current number):

1819369361361181936936136110106418193618193610101193613616425641819361819369
3613611936136164251641819361819361193613616425641819361819361641819361819361
361616418193610193613616425641819361819361193613616425181936

-> After squaring digits:

1641819368193619361164181936819361936110103616164181936164181936101011819361
9361361642536161641819361641819368193619361181936193613616425136161641819361
6418193611819361936136164253616164181936164181936136161641819361641819361936
1361361616418193610181936193613616425361616418193616418193611819361936136164
25164181936

-> After summing groups of 3:

111018181018813131518138110131313131372101813101311101313131313151813818
1018101181013131313131381810181011101311101811101810811181011181018101081118
1091316101012148111810111810101813101311111018

-> After summing groups of 3:

3910912514122575710954345759101291029575717293439399393910110101010231339392
122539

Step 67 (current number):

3910912514122575710954345759101291029575717293439399393910110101010231339392
122539

-> After squaring digits:

9811081142511614425492549108125169162549258110148110481254925491494819169819
8181981981101101010104919981981414425981

Step 68 (current number):

9811081142511614425492549108125169162549258110148110481254925491494819169819
8181981981101101010104919981981414425981

-> After squaring digits:

8164110641116425113611616425168142516811064142513681136425168142564110116641
1016641425168142516811168116641811368164181641641816418164110110101010168118
18164181641161161642581641

Step 69 (current number):

8164110641116425113611616425168142516811064142513681136425168142564110116641
1016641425168142516811168116641811368164181641641816418164110110101010168118
18164181641161161642581641

-> After squaring digits:

6413616110361611136164251193611361361642513664116425136641103616116425193664
1193616425136641164253616110113636161101363616116425136641164251366411136641
1363616164119366413616164136161361616413616164136161101101010101366411641641
361616413616113611361361642564136161

-> After summing groups of 3:

1110898513111110101011815612916210812151561811815612148210108415811815612916
31561213111115813111081013813111082212156111110138135810101215108

-> After summing groups of 3:

317145312101212998121010127111131410121210126371239412310512321299289

-> After summing groups of 3: 1110625264110355439111474862019

Step 70 (current number): 1110625264110355439111474862019

-> After squaring digits: 111036425436161109252516981111164916643640181

-> After summing groups of 3: 3911138109121831116131010

Step 71 (current number): 3911138109121831116131010

-> After squaring digits: 981111964108114164911136191010

-> After summing groups of 3: 1831996111110111

Step 72 (current number): 1831996111110111

-> After squaring digits: 16491818136111110111

Step 73 (current number): 16491818136111110111

-> After squaring digits: 13616811641641936111110111

Step 74 (current number): 13616811641641936111110111

-> After squaring digits: 19361366411361613616181936111110111

Step 75 (current number): 19361366411361613616181936111110111

-> After squaring digits: 18193619363616119361361936136164181936111110111

Step 76 (current number): 18193619363616119361361936136164181936111110111

-> After squaring digits:

16418193618193693613611819361936181936193613616164181936111110111

Step 77 (current number):

16418193618193693613611819361936181936193613616164181936111110111

-> After squaring digits:

1361616418193616418193681936193611641819361819361641819361819361936136136161
64181936111110111

-> After summing groups of 3:

1081118101118171316101110181018111018101813101013111018323

Step 78 (current number):

1081118101118171316101110181018111018101813101013111018323

-> After squaring digits:

10641116410111641491913610111016410164111016410164191010191110164949

Step 79 (current number):

10641116410111641491913610111016410164111016410164191010191110164949

-> After squaring digits:

1036161113616101113616116811811936101110136161013616111013616101361618110101
81111013616811681

-> After summing groups of 3: 4133107310815101373413210821071081011034131015

Step 80 (current number): 4133107310815101373413210821071081011034131015

-> After squaring digits: 161991049910641251019499161941064410491064101109161910125

-> After summing groups of 3: 8191310118102281410514102108108

Step 81 (current number): 8191310118102281410514102108108

-> After squaring digits: 6418119101164104464116102511610410641064

Step 82 (current number): 6418119101164104464116102511610410641064

-> After squaring digits:

3616164118110113616101616361611361042511361016103616103616

Step 83 (current number):

3616164118110113616101616361611361042511361016103616103616

-> After squaring digits:

936136136161164110119361361013613693613611936101642511936101361093613610936136

-> After summing groups of 3: 18101081121110104101810818212718210121071810

Step 84 (current number): 18101081121110104101810818212718210121071810

-> After squaring digits: 164101064114111010161016410641644144916441014104916410

-> After summing groups of 3: 112106318751114916955165

-> After summing groups of 3: 4712136161912

Step 85 (current number): 4712136161912

-> After squaring digits: 164914193613618114

-> After summing groups of 3: 11141310156

Step 86 (current number): 11141310156

-> After squaring digits: 11116191012536

Step 87 (current number): 11116191012536

-> After squaring digits: 111136181101425936

-> After summing groups of 3: 3101021118

Step 88 (current number): 3101021118

-> After squaring digits: 91010411164

Step 89 (current number): 91010411164

-> After squaring digits: 811010161113616

-> After summing groups of 3: 1018513

Step 90 (current number): 1018513

-> After squaring digits: 101642519

-> After summing groups of 3: 21215

Step 91 (current number): 21215

-> After squaring digits: 414125

-> After summing groups of 3: 98

Step 92 (current number): 98

-> After squaring digits: 8164

Step 93 (current number): 8164

-> After squaring digits: 6413616

Step 94 (current number): 6413616

-> After squaring digits: 36161936136

Step 95 (current number): 36161936136

-> After squaring digits: 9361361819361936

Step 96 (current number): 9361361819361936

-> After squaring digits: 819361936164181936181936

-> After summing groups of 3: 1810181110181018

Step 97 (current number): 1810181110181018

-> After squaring digits: 16410164111016410164

Step 98 (current number): 16410164111016410164

-> After squaring digits: 1361610136161110136161013616

Step 99 (current number): 1361610136161110136161013616

-> After squaring digits: 193613610193613611101936136101936136

-> After summing groups of 3: 13107131082181021810

Step 100 (current number): 13107131082181021810

-> After squaring digits: 191049191064416410416410

-> After summing groups of 3: 11131110115115

Step 101 (current number): 11131110115115

-> After squaring digits: 1119111011251125

Step 102 (current number): 1119111011251125

-> After squaring digits: 1118111101142511425

Step 103 (current number): 1118111101142511425

-> After squaring digits: 111641111011164251116425

-> After summing groups of 3: 31132118811

Step 104 (current number): 31132118811

-> After squaring digits: 9119411646411

Step 105 (current number): 9119411646411

-> After squaring digits: 81118116113616361611

Step 106 (current number): 81118116113616361611

-> After squaring digits: 641116411361193613693613611

-> After summing groups of 3: 118610131018108

-> After summing groups of 3: 107599

-> After summing groups of 3: 823

-> After summing groups of 3: 13

Step 107 (current number): 13

-> After squaring digits: 19

Step 108 (current number): 19

-> After squaring digits: 181

-> After summing groups of 3: 10

Step 109 (current number): 10

-> After squaring digits: 10

Brutal stable number reached: 10

2.2 The number 855544410

This number takes 22 steps to reach the stable number 1011 as illustrated below:

Enter the starting number: 855544410

Step 1 (current number): 855544410

-> After squaring digits: 6425252516161610

Step 2 (current number): 6425252516161610

-> After squaring digits: 361642542542513613613610

-> After summing groups of 3: 10121111910107

Step 3 (current number): 10121111910107

-> After squaring digits: 1014111181101049

Step 4 (current number): 1014111181101049

-> After squaring digits: 10116111164110101681

Step 5 (current number): 10116111164110101681

-> After squaring digits: 1011361111361611010136641

Step 6 (current number): 1011361111361611010136641

-> After squaring digits: 1011936111193613611010193636161

Step 7 (current number): 1011936111193613611010193636161

-> After squaring digits: 1011819361111819361936110101819369361361

Step 8 (current number): 1011819361111819361936110101819369361361

-> After squaring digits: 101164181936111164181936181936110101641819368193619361

-> After summing groups of 3: 21110183111018101822111817131610

Step 9 (current number): 21110183111018101822111817131610

-> After squaring digits: 41110164911101641016444111641491913610

Step 10 (current number): 41110164911101641016444111641491913610

-> After squaring digits: 161110136168111101361610136161616111361611681181193610

-> After summing groups of 3: 821015321071081331081510137

-> After summing groups of 3: 11668979611

Step 11 (current number): 11668979611

-> After squaring digits: 113636648149813611

-> After summing groups of 3: 5151814128

Step 12 (current number): 5151814128

-> After squaring digits: 251251641161464

-> After summing groups of 3: 8811814

Step 13 (current number): 8811814

-> After squaring digits: 64641164116

Step 14 (current number): 64641164116

-> After squaring digits: 361636161136161136

-> After summing groups of 3: 1015810810

Step 15 (current number): 1015810810

-> After squaring digits: 1012564106410

Step 16 (current number): 1012564106410

-> After squaring digits: 101425361610361610

-> After summing groups of 3: 211107107

-> After summing groups of 3: 488

-> After summing groups of 3: 20

Step 17 (current number): 20

-> After squaring digits: 40

Step 18 (current number): 40

-> After squaring digits: 160

-> After summing groups of 3: 7

Step 19 (current number): 7

-> After squaring digits: 49

Step 20 (current number): 49

-> After squaring digits: 1681

Step 21 (current number): 1681

-> After squaring digits: 136641

-> After summing groups of 3: 1011

Step 22 (current number): 1011

-> After squaring digits: 1011

Brutal stable number reached: 1011

2.3 The number 88888

This number takes 13 steps to reach the stable number 1111111111 as illustrated below:

Enter the starting number: 88888

Step 1 (current number): 88888

-> After squaring digits: 6464646464

Step 2 (current number): 6464646464

-> After squaring digits: 3616361636163616

Step 3 (current number): 3616361636163616

-> After squaring digits: 936136936136936136936136936136

-> After summing groups of 3: 18101810181018101810

Step 4 (current number): 18101810181018101810

-> After squaring digits: 1641016410164101641016410

Step 5 (current number): 1641016410164101641016410

-> After squaring digits: 13616101361610136161013616101361610

Step 6 (current number): 13616101361610136161013616101361610

-> After squaring digits: 193613610193613610193613610193613610193613610

-> After summing groups of 3: 131071310713107131071310713107

Step 7 (current number): 131071310713107131071310713107

-> After squaring digits: 191049191049191049191049191049191049

-> After summing groups of 3: 111311131113111311131113

Step 8 (current number): 111311131113111311131113

-> After squaring digits: 111911191119111911191119

Step 9 (current number): 11191119111911191119

-> After squaring digits: 1118111181111811118111181

Step 10 (current number): 1118111181111811118111181

-> After squaring digits: 111641111641111641111641111641

-> After summing groups of 3: 311311311311311

-> After summing groups of 3: 55555

Step 11 (current number): 55555

-> After squaring digits: 2525252525

Step 12 (current number): 2525252525

-> After squaring digits: 425425425425425

-> After summing groups of 3: 1111111111

Step 13 (current number): 1111111111

-> After squaring digits: 1111111111

Brutal stable number reached: 1111111111

3.Real World Implications

3.1 Cryptography

The system's trajectories act as a digit level one way function. Inputs explode into massive non-linear sequences before collapsing into fixed points making prediction or reverse engineering computationally infeasible. This property is suitable for hashing schemes, key generation or secure encoding mechanisms providing robust resistance to attacks due to the systems extreme combinatorial complexity.

3.2 Artificial Intelligence and Pattern Recognition

Each integer produces a unique trajectory that can be treated as a high dimensional feature vector. Machine learning algorithms can classify numbers by attractor basins, detect hidden structures or identify convergence patterns. This creates a new paradigm for integer-based feature extraction where deterministic chaos informs predictive modeling.

3.3 Pseudorandom Number Generation (PRNG)

The system produces gigantic intermediate sequences with high entropy. Controlled extraction of digits can yield long period, nonlinear pseudo random streams, deterministic yet unpredictable without full iteration. This property makes it ideal for simulation, stochastic modelling and cryptographically secure PRNGs.

3.4 Discrete Dynamical System Modelling

This system serves as a sandbox for studying deterministic chaos, basins of attraction and infinite attractors in discrete integer spaces. It can model iterative processes in population dynamics, resource growth or cascading systems providing insight into complex systems where growth and collapse coexist.

3.5 Educational and Research Tool

This system is highly visual and computationally tangible. It demonstrates chaos, stabilization and attraction structures in a manner that traditional integer problems cannot. Students and researchers can explore convergence patterns, explosive growth and fixed-point analysis directly through computation.

Conclusion

This work introduces a novel class of digit based dynamical systems governed by iterative digit squaring and group of three compression which exhibits unprecedented explosive behavior coupled with multiple finite attractors. Unlike classical systems such as Collatz or Kaprekar routines which either converge slowly or are constrained by fixed digit lengths, the system presented here demonstrates:

[1] Exponential digit growth in intermediate iterations producing states billions of times larger than initial inputs.

[2] Deterministic collapse to fixed points including 11, 10, 1011, 1010, 1110 and selective hyper ones establishing a new class of infinite attractors.

The system is fully deterministic yet explosively chaotic creating a digit universe simulator with properties not observed in prior integer iteration systems. Its combination of computational explosion and stabilization offers fertile ground for further research in integer dynamics, number theory and computational modelling.

This study lays the groundwork for future applications in pattern recognition, cryptography and mathematical experimentation while simultaneously challenging the current understanding of iterative integer systems. The results suggest a 21st century super Collatz, a system capable of demonstrating deterministic chaos on a scale far beyond classical analogues.